

SRIE

**Project 1 - Sycophancy as a set
of constructs**

BILEL HATMI, SUPERVISION

EIGHT WEEKS

4 August to 28 September
2026

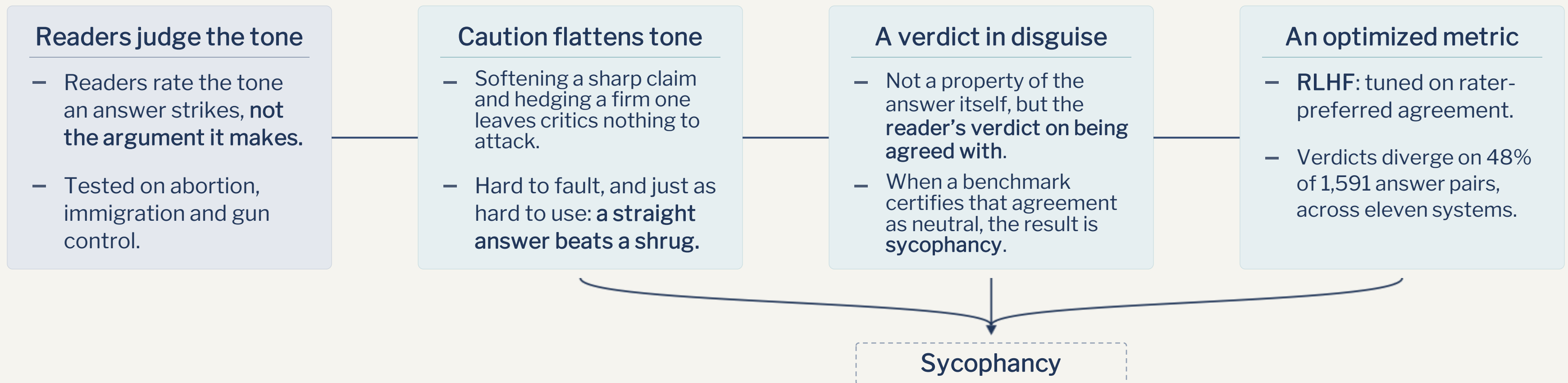


The project

Turning sycophancy into valid constructs for LLMs

Trust in a model is decided by its tone

One person talked into believing something false by a chatbot is a small problem, and a familiar one for psychology to study. The problem changes scale when a whole population forms its sense of what is fair, on health, on law, on a live political fight, from a handful of systems built the same way.



An answer that agrees tends to come out soft and considered; one that pushes back rarely gets the same treatment.

The cue holds between people and breaks on a model

The human account fails, and five causes take its place

- 1 Neither the cue nor the account behind it carries over to a model.
- 2 Five separately caused properties: procedural validity, epistemic validity, direction, conviction, dispersion.



Five mechanisms mistaken for one

- When a model consistently tells people what they want to hear, no single trait explains it: five act at once.
- The usual word names it as one trait, and so assumes one cause.
 - Here, there may be no single cause.

Each piece has to be traced back to whichever stage produced it, and counted rather than felt.

Four quantities, combined into a profile with its own regime

Each of the four has its own cause and failure mode

	PSYCHOLOGICAL CONSTRUCT	POTENTIAL CAUSES	TENDENCIES	FAILURE MODE
Trajectory validity	Reasoning ability, adaptability	Capability training, dataset quality	Locally sound, incomplete	Confident but unfounded claims
Direction	Sycophancy (narrow sense)	RLHF; preference-model tuning	Tracks the asker's position	Flips with who's asking
Conviction	Confidence, assertiveness	RLHF plus safety tuning	Decouples from actual validity	Confidence without grounding
Dispersion	Consistency across a population	Deliberate variance reduction	Converges on one safe answer	Narrows the views people meet

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UNFOLDS
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Named as one profile, but how it moves stays untested

- Taken side by side, they raise a question none of them raises alone: how they move together, and what links them.
- The four, measured on one case, make up a model's profile.

A regime is the range of conditions over which that profile, and the coupling between its parts, stays roughly steady.

Two instruments run across five conditions

Two instruments, one cheap and one costly

- Both instruments aim at the same four quantities, but only one is cheap to build.
- **Internals:** linear probes and persona vectors are costly to get right, and what they report may be nothing more than the model's own output variance.
- **Behaviour:** what the model does, measured from outside, without that cost.
- Neither instrument alone can confirm that its own reading is right.

Five conditions define the sweep

- Each run varies:
 - **Claim subjectivity**, how contested it is
 - **Point of view**, first or third person
 - **Persona**, which role the model has been given
 - **Conversational history**, how long the exchange has run
 - **Sampling**, how the answer gets drawn

Running both is a **cross-check**: two readings of the same quantity, each one a test of the other.

The project’s next phase **rests on three open questions**

Coupling stability	Regime shape and grain	Intervention
While persistent disorder in the coupling between the four would point toward specialised systems for education, law, and health, behaviour settling instead into one general regime would let the general-purpose case survive.	Generalizability theory, SEMs, Clustering analysis... are methods named for settling, group by group whether sycophancy forms a coherent and more fundamental structure.	Within a regime set by claim subjectivity, point of view, persona, conversational history, and sampling, the aim is to find what holds a model at a chosen equilibrium, then steer it with a steering vector or a targeted fine-tune.

Flatten the useful spread along with the harmful part, and models stop being worth comparing.



Supervising the students

This program is meant to be more than just a research project

More than a project

Four things get built alongside the research, and they are on the timetable like everything else.

Method



- 3 cross-cutting themes, everyone in the room regardless of project: **literature review, reading and writing a paper, designing an experiment.**
- Turning a qualitative question into a **valid statistical design**, and **criticising an experiment while reading it.**

Tools



- **Claude Code** across the whole range, **from literature to code, wired into a single connected ecosystem.**
- Using **organisational tools like Jira, Drive and Slack** to improve efficiency and communication

Communication



- **Presenting technical work** in a structured way
- **Asking the right questions** at the right moment to **encourage creativity**, and reacting appropriately when knowledge is missing.

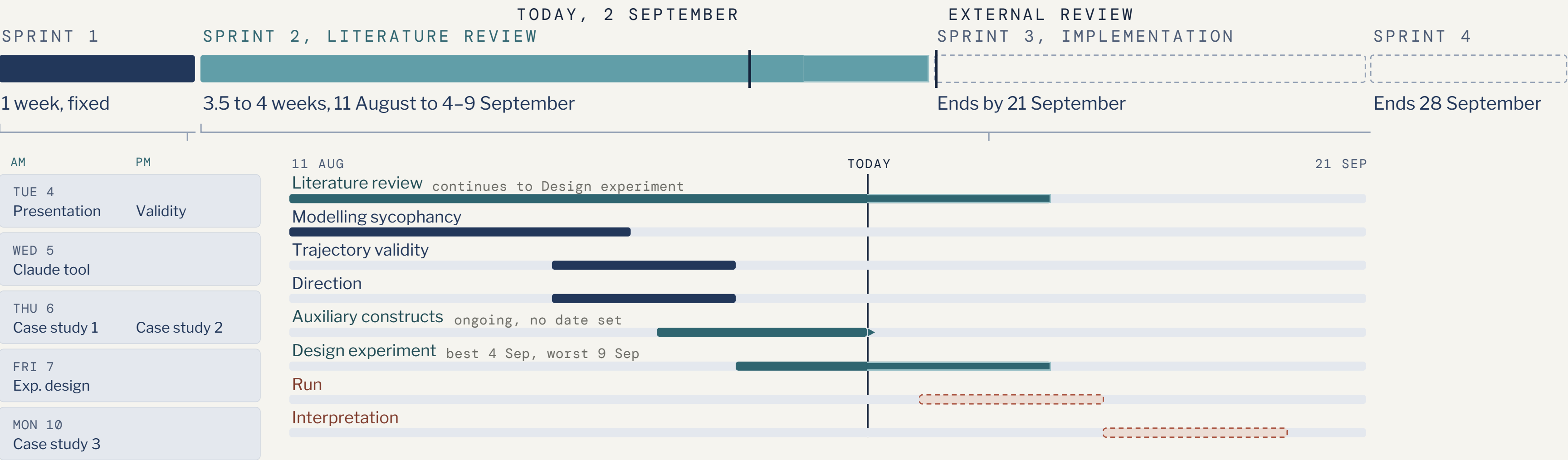
Yourself



- **A time to get to know yourself and work more efficiently** (Productivity, Rest, Learning, feedback culture, fear of comparison, perfectionism...). These are pain points common to highly demanding work such as research.

The team is measuring what AI does to the people who use it, so it is unusually well placed **to study its own use of AI tools.**
The next page sets out the eight-week calendar.

The roadmap



Before the empirical work begins, the students present the design to a member of the AI safety mathematics community. Vardhan Kumar Ray, formerly a fellow at Iliad and SPAR.

How the stream runs

Three levels, each carrying a different register of work. No level does the job of another.



The whole cohort for **what is deemed useful to everyone**, the group meeting to **keep the needs and ambitions of each stream coherent**, the individual meeting to **delve deeper into technical issues**.

Core bibliography for now

1. What establishes the problem

Towards Understanding Sycophancy in Language Models.

Sharma et al., 2024. ICLR. arXiv:2310.13548

Sycophancy Is Not One Thing: Causal Separation of Sycophantic Behaviors in LLMs.

Vennemeyer et al., 2025. arXiv:2509.21305

2. Epistemic vigilance and trajectory validity

Epistemic Vigilance.

Sperber et al., 2010. Mind and Language, 25(4)

Accommodation and Epistemic Vigilance: A Pragmatic Account of Why LLMs Fail to Challenge Harmful Beliefs.

Cheng, Hawkins and Jurafsky, 2026. arXiv:2601.04435

Rational Metareasoning for Large Language Models.

De Sabbata et al., 2024. arXiv:2410.05563

When Truth Is Overridden: Uncovering the Internal Origins of Sycophancy in Large Language Models.

Wang et al., 2025. arXiv:2508.02087

3. Instruments and methodology

BASIL: Bayesian Assessment of Sycophancy in LLMs.

Atwell et al., 2025. arXiv:2508.16846

Let's Verify Step by Step.

Lightman et al., 2023. arXiv:2305.20050